

Structural MRI Analysis Manual

FreeSurfer 7.4.1: cortical reconstruction, morphometry, ROI tables, quality control, and group analysis

Laboratory workflow - English edition - July 2026

Audience: students processing research-grade T1-weighted MRI on a shared Linux server.

Primary software: FreeSurfer 7.4.1, Bash, Python 3.11, pandas, and statsmodels.

Starting point: a valid BIDS dataset or a completed fMRIPrep derivatives directory.

Deliverables: subject-level recon-all outputs, QC records, ROI tables, and group-ready cortical thickness data.

All example project paths use /your_folder/.

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1. Purpose and analysis boundaries

This manual describes a reproducible cross-sectional structural MRI workflow for research T1-weighted images. The main subject-level operation is FreeSurfer recon-all, which performs intensity normalization, skull stripping, subcortical segmentation, white and pial surface reconstruction, cortical parcellation, and morphometric measurement. The manual then converts those outputs into regional tables and surface-based group analyses.

Scope limitation: This is a cross-sectional workflow. Longitudinal studies should use the dedicated FreeSurfer longitudinal stream because independently processing each visit can reduce sensitivity and introduce avoidable variability.

- Primary outcomes: cortical thickness, surface area, cortical gray-matter volume, and subcortical volumes.
- Primary QC: visual inspection of skull stripping, white surfaces, pial surfaces, and segmentation labels.
- Primary group approaches: ROI-wise regression and vertex-wise general linear models on fsaverage.
- Do not mix FreeSurfer versions within a study unless a formal harmonization plan has been defined.

2. Workflow overview

1. Confirm a valid T1w image and a stable participant label.
2. Select one FreeSurfer output tree: reuse completed fMRIPrep surfaces or run standalone FreeSurfer 7.4.1.
3. Run recon-all and preserve logs.
4. Perform subject-level QC before extracting statistics.
5. Export regional measurements with asegstats2table and aparcstats2table.
6. Merge morphometry with participants.tsv and fit prespecified statistical models.
7. For vertex-wise analysis, resample subject maps to fsaverage, smooth once, fit the GLM, and correct for multiple comparisons.

Version pinning: The server workflow in this manual intentionally uses FreeSurfer 7.4.1. The latest major FreeSurfer releases may change algorithms and outputs; reproducibility is improved by recording and fixing the exact version.

3. Required data and project layout

Use a predictable directory structure. The following layout separates raw BIDS data, derivatives, code, logs, and analysis tables.

```
/your_folder/
├── BIDS/
│   ├── dataset_description.json
│   ├── participants.tsv
│   ├── sub-GAB01/
│   │   └── anat/
│   │       └── sub-GAB01_T1w.nii.gz
├── code/
│   └── structural/
├── derivatives/
│   ├── fmripred-25.2.5/
│   ├── freesurfer-7.4.1/
│   ├── structural-stats/
│   └── structural-surface-glm/
├── logs/
└── freesurfer/
```

Permissions: On a shared server, create a laboratory Unix group and give the project directories group read/write permission. Do not make protected participant data world-readable.

4. Software and environment setup

4.1 FreeSurfer environment

FreeSurfer is already installed on the server at /usr/local/freesurfer/7.4.1. It is not installed inside the Python conda environment.

```
export FREESURFER_HOME=/usr/local/freesurfer/7.4.1
export FS_LICENSE=/usr/local/freesurfer/7.4.1/license.txt
source "$FREESURFER_HOME/SetUpFreeSurfer.sh"
```

```
recon-all -version
which recon-all
ls -l "$FS_LICENSE"
```

A successful setup prints the FreeSurfer version, the recon-all executable path, and a readable license file.

4.2 Python environment for tables and statistics

```
cd /your_folder/code/structural
conda env create -f environment.yml
conda activate structural_stats
python -m ipykernel install --user --name structural_stats --display-name "Python (structural_stats)"
python -c "import pandas, statsmodels; print('Python environment OK')"
```

Environment separation: FreeSurfer performs image reconstruction. The structural_stats conda environment is only for tabular analysis, plots, and notebooks. Keeping these layers separate reduces dependency conflicts.

5. Decide whether to reuse fMRIPrep FreeSurfer outputs

fMRIPrep can run FreeSurfer and store a complete subject directory under its derivatives tree. Reusing a completed reconstruction avoids running recon-all twice.

5.1 Check for an existing subject

```
find /your_folder/derivatives/fmriprep-25.2.5/sourcedata/freesurfer/sub-GAB01 -maxdepth 2 -type f | head -30
cat /your_folder/derivatives/fmriprep-25.2.5/sourcedata/freesurfer/sub-GAB01/scripts/recon-all.done 2>/dev/null ||
true
tail -n 20 /your_folder/derivatives/fmriprep-25.2.5/sourcedata/freesurfer/sub-GAB01/scripts/recon-all.log
```

5.2 Reuse rule

- Reuse the fMRIPrep subject only when recon-all completed successfully and all participants were processed with the same fMRIPrep and FreeSurfer versions.
- Set SUBJECTS_DIR to the fMRIPrep FreeSurfer directory for all downstream FreeSurfer commands.
- Do not copy only selected files from one FreeSurfer tree into another.
- If the fMRIPrep reconstruction is missing, incomplete, or was generated with an incompatible protocol, run the standalone workflow below.

```
export SUBJECTS_DIR=/your_folder/derivatives/fmriprep-25.2.5/sourcedata/freesurfer
ls "$SUBJECTS_DIR/sub-GAB01"
```

6. Verify T1w inputs and participant identifiers

```
SUBJECT=GAB01
T1w=/your_folder/BIDS/sub-{$SUBJECT}/anat/sub-{$SUBJECT}_T1w.nii.gz

ls -lh "$T1w"
mri_info "$T1w" | head -40
awk 'NR==1 || $1=="sub-'"${SUBJECT}"' /your_folder/BIDS/participants.tsv'
```

- The subject label supplied to scripts excludes the sub- prefix: use GAB01, not sub-GAB01.
- The anatomical filename must match the BIDS participant label exactly.
- Inspect image orientation before processing. Gross left-right mistakes cannot be repaired by statistical analysis.
- For repeated T1w acquisitions in the same session, define a documented combination strategy rather than silently choosing one file.

7. Run recon-all for one participant

Copy the supplied structural scripts to /your_folder/code/structural and run the single-subject script.

```
mkdir -p /your_folder/code/structural
cd /your_folder/code/structural

./01_check_freesurfer.sh
./02_run_recon_all.sh GAB01
```

The script resolves the default input and output paths as follows:

Item	Path
Input T1w	/your_folder/BIDS/sub-GAB01/anat/sub-GAB01_T1w.nii.gz

Item	Path
FreeSurfer subject	/your_folder/derivatives/freesurfer-7.4.1/sub-GAB01
External log	/your_folder/logs/freesurfer/recon-all_sub-GAB01.log
Internal log	/your_folder/derivatives/freesurfer-7.4.1/sub-GAB01/scripts/recon-all.log

7.1 Direct command equivalent

```
export FREESURFER_HOME=/usr/local/freesurfer/7.4.1
export FS_LICENSE=$FREESURFER_HOME/license.txt
export SUBJECTS_DIR=/your_folder/derivatives/freesurfer-7.4.1
source "$FREESURFER_HOME/SetUpFreeSurfer.sh"
mkdir -p "$SUBJECTS_DIR" /your_folder/logs/freesurfer

recon-all -subject sub-GAB01 -i /your_folder/BIDS/sub-GAB01/anat/sub-GAB01_T1w.nii.gz -all 2>&1 | tee
/your_folder/logs/freesurfer/recon-all_sub-GAB01.log
```

Do not overwrite: After recon-all has imported the anatomical image, do not rerun the same subject with a different -i input in the same subject directory. Use a new output subject name or remove the incomplete directory only after preserving logs and confirming the decision.

8. Use tmux and monitor long jobs

recon-all can take many hours. Run it inside tmux so the job survives an interrupted SSH or VS Code connection.

```
tmux new -s recon_GAB01
cd /your_folder/code/structural
./02_run_recon_all.sh GAB01
```

Detach without stopping the job: press Ctrl+b, release the keys, then press d.

```
tmux ls
tmux attach -t recon_GAB01

tail -f /your_folder/logs/freesurfer/recon-all_sub-GAB01.log
```

9. Batch processing

The batch script reads the first column of participants.tsv. Sequential processing is the safe default because multiple recon-all jobs can saturate memory and storage I/O.

```
cd /your_folder/code/structural
MAX_JOBS=1 ./03_batch_recon_all.sh /your_folder/BIDS/participants.tsv
```

Parallel jobs: Increase MAX_JOBS only after checking free -h, nproc, and disk throughput. A shared server can become slower when too many recon-all jobs compete for RAM and I/O.

10. Determine completion and restart safely

```
cd /your_folder/code/structural
./04_check_recon_status.sh GAB01

ls /your_folder/derivatives/freesurfer-7.4.1/sub-GAB01/scripts/
tail -n 50 /your_folder/derivatives/freesurfer-7.4.1/sub-GAB01/scripts/recon-all.log
```

Marker	Meaning
recon-all.done	The full requested workflow completed successfully.
recon-all.error	The workflow failed. Read the internal log before retrying.
Log only, no marker	The job is still running, was interrupted, or did not reach a clean exit.

After correcting the cause of an interrupted run, the usual restart is:

```
recon-all -subject sub-GAB01 -all
```

Restart principle: recon-all is stage-aware and normally resumes missing steps. Do not delete the whole subject directory merely because a job stopped; first inspect the log and error marker.

11. Quality control

A successful exit code is necessary but not sufficient. Every subject must receive visual QC before inclusion in morphometric analysis.

11.1 Automated checks

```
SUBJECTS_DIR=/your_folder/derivatives/freesurfer-7.4.1
SUBJ=sub-GAB01

ls "$SUBJECTS_DIR/$SUBJ/scripts/recon-all.done"
ls "$SUBJECTS_DIR/$SUBJ/mri/aseg.mgz"
ls "$SUBJECTS_DIR/$SUBJ/surf/lh.white" "$SUBJECTS_DIR/$SUBJ/surf/lh.pial"
grep -iE "error|failed|warning" "$SUBJECTS_DIR/$SUBJ/scripts/recon-all.log" | tail -30
```

11.2 Visual inspection with Freeview

```
freeview -v /your_folder/derivatives/freesurfer-7.4.1/sub-GAB01/mri/T1.mgz
/your_folder/derivatives/freesurfer-7.4.1/sub-GAB01/mri/aseg.mgz:colormap=lut:opacity=0.35 -f
/your_folder/derivatives/freesurfer-7.4.1/sub-GAB01/surf/lh.white:edgecolor=blue
/your_folder/derivatives/freesurfer-7.4.1/sub-GAB01/surf/lh.pial:edgecolor=red
/your_folder/derivatives/freesurfer-7.4.1/sub-GAB01/surf/rh.white:edgecolor=blue
/your_folder/derivatives/freesurfer-7.4.1/sub-GAB01/surf/rh.pial:edgecolor=red
```

- Skull stripping: brain tissue should be retained and dura, skull, and neck should be excluded.
- White surface: should track the gray-white boundary without cutting through cortex or including deep gray matter.
- Pial surface: should follow the outer cortical boundary without capturing dura or large vessels.
- Aseg labels: ventricles, thalamus, caudate, putamen, pallidum, hippocampus, and amygdala should be anatomically plausible.
- Temporal poles and orbitofrontal cortex: inspect carefully because susceptibility and dura can cause surface errors.
- Record pass, minor issue, major issue, exclusion, and any manual edits in a study QC table.

Manual edits: Manual control points, white-matter edits, and pial edits must follow a written lab protocol and be performed blind to group when possible. The same criteria must be applied to all participants.

12. Understand the output tree

Directory/file	Purpose
mri/orig.mgz	Conformed input volume used by FreeSurfer.
mri/brainmask.mgz	Skull-stripped anatomical volume.
mri/aseg.mgz	Automated subcortical segmentation.
surf/lh.white, rh.white	White-matter cortical surfaces.
surf/lh.pial, rh.pial	Pial cortical surfaces.
surf/lh.thickness, rh.thickness	Vertex-wise cortical thickness.
label/lh.aparc.annot, rh.aparc.annot	Desikan-Killiany cortical parcellation.
stats/aseg.stats	Subcortical volume statistics and estimated total intracranial volume.
stats/lh.aparc.stats, rh.aparc.stats	Regional cortical thickness, area, and volume.
scripts/recon-all.log	Complete processing log and version information.

13. Extract regional morphometry tables

After QC, create one row per participant and one column per anatomical region.

```
cd /your_folder/code/structural
./05_extract_roi_tables.sh /your_folder/BIDS/participants.tsv

ls -lh /your_folder/derivatives/structural-stats/
head -2 /your_folder/derivatives/structural-stats/aseg_volume.tsv
head -2 /your_folder/derivatives/structural-stats/lh_aparc_thickness.tsv
```

Output table	Measurement
aseg_volume.tsv	Subcortical and global volumes, including eTIV-related fields.
lh_aparc_thickness.tsv / rh_aparc_thickness.tsv	Mean cortical thickness by Desikan-Killiany region.

Output table	Measurement
lh_aparc_area.tsv / rh_aparc_area.tsv	Cortical surface area by region.
lh_aparc_volume.tsv / rh_aparc_volume.tsv	Cortical gray-matter volume by region.

QC first: Do not generate a final statistical table and then forget which subjects failed surface QC. Maintain an explicit inclusion column and apply it before modeling.

14. Build analysis-ready ROI datasets

The first column of each FreeSurfer table contains the subject identifier. Standardize it before merging with participants.tsv.

```
conda activate structural_stats
python - <<'PY'
import pandas as pd

roi = pd.read_csv('/your_folder/derivatives/structural-stats/aseg_volume.tsv', sep=' ')
roi = roi.rename(columns={roi.columns[0]: 'participant_id'})
roi['participant_id'] = roi['participant_id'].str.replace(r'^sub-', '', regex=True)

participants = pd.read_csv('/your_folder/BIDS/participants.tsv', sep=' ')
participants['participant_id'] = participants['participant_id'].str.replace(r'^sub-', '', regex=True)

data = participants.merge(roi, on='participant_id', how='inner', validate='one_to_one')
print(data.shape)
print(data[['participant_id', 'group', 'age', 'sex']].head())
data.to_csv('/your_folder/derivatives/structural-stats/aseg_analysis_ready.tsv', sep=' ', index=False)
PY
```

- Verify that the merged row count equals the expected number of included participants.
- Check duplicates before using validate="one_to_one".
- Keep raw exported tables unchanged and write cleaned tables as new files.
- Document units: thickness in millimeters; area in square millimeters; volume in cubic millimeters.

15. ROI-level statistical modeling

The supplied Python script fits one OLS model per region and applies Benjamini-Hochberg FDR across the tested ROIs.

```
conda activate structural_stats
python /your_folder/code/structural/analyze_roi_stats.py --roi-table
/your_folder/derivatives/structural-stats/lh_aparc_thickness.tsv --participants /your_folder/BIDS/participants.tsv
--predictor group --covariates age sex --output
/your_folder/derivatives/structural-stats/lh_thickness_group_glm.tsv
```

15.1 Intracranial volume

- For regional volume outcomes, include estimated total intracranial volume (eTIV) as a covariate when scientifically appropriate.
- Do not automatically divide every regional volume by eTIV. Ratio normalization changes the estimand and can introduce bias.
- Cortical thickness is generally not scaled by eTIV, while surface area and volume often require head-size consideration.
- Prespecify age, sex, scanner/site, and other covariates based on the design, not on which model produces significance.

15.2 Example formula

```
regional_volume ~ group + age + sex + eTIV
cortical_thickness ~ group + age + sex
```

Categorical coding: The coefficient name for a categorical predictor depends on the reference level. Inspect the model design and report the actual contrast, not merely the word group.

16. Vertex-wise cortical thickness analysis

Vertex-wise analysis tests cortical thickness at each fsaverage vertex. The design matrix, smoothing kernel, and contrast must be fixed before viewing statistical maps.

16.1 Create an FSGD file

```
GroupDescriptorFile 1
Title GroupThickness
Class Control
Class Patient
Variables age
Input sub-GAB01 Control 24
Input sub-GAB02 Control 29
Input sub-GAB03 Patient 27
Input sub-GAB04 Patient 31
```

Save the file as `/your_folder/code/structural/group_thickness.fsgd`. The Input subject names must exactly match folders in `SUBJECTS_DIR`.

16.2 Prepare common-space thickness maps

```
cd /your_folder/code/structural
./06_prepare_surface_glm.sh /your_folder/code/structural/group_thickness.fsgd 10
```

16.3 Define a contrast

For a two-class FSGD using DODS or DOSS, the contrast dimensions depend on the generated design. Verify the design before applying a contrast. A simple group contrast file might contain:

```
1 -1 0
```

Contrast verification: Do not copy a contrast matrix from another study without checking the FSGD design columns. Use `mri_glmfit --checkopts` and inspect `X.mat` or the generated design information.

16.4 Fit the GLM

```
./07_run_surface_glm.sh /your_folder/code/structural/group_thickness.fsgd
/your_folder/code/structural/patient_minus_control.mtx 10
```

16.5 Inspect maps

```
freeview -f $SUBJECTS_DIR/fsaverage/surf/lh.inflated:overlay=/your_folder/derivatives/structural-surface-glm/
lh.thickness.sm10.glmDir/patient_minus_control/sig.mgh:annot=aparc.annot
```

17. Multiple comparisons and reporting

- ROI analyses: correct across all ROIs in the prespecified family, commonly with FDR or Bonferroni.
- Vertex-wise analyses: use a surface-appropriate correction method such as permutation/Monte Carlo cluster correction or vertex-wise FDR, according to the analysis plan.
- Correct across both hemispheres if they belong to the same inferential family.
- Report FreeSurfer version, atlas, smoothing FWHM, design matrix, contrast, covariates, exclusion rules, and correction method.
- Provide effect sizes and confidence intervals in addition to corrected p-values.

Exploratory maps: Clearly label uncorrected surface maps as exploratory. Do not present an uncorrected vertex-wise threshold as confirmatory evidence.

18. Optional hippocampal subfield segmentation

FreeSurfer includes hippocampal subfield segmentation tools. Run them only after a completed recon-all and use the same tool version for every participant.

```
export SUBJECTS_DIR=/your_folder/derivatives/freesurfer-7.4.1
source /usr/local/freesurfer/7.4.1/SetUpFreeSurfer.sh
segmentHA_T1.sh sub-GAB01
```

- Inspect the command log and generated segmentation before using volumes.
- Treat subfield boundaries as model-based estimates, especially for standard-resolution T1w data.

- Avoid selecting only the subfields that show favorable results; define the hypothesis and correction family first.

19. Reproducibility and provenance

```
mkdir -p /your_folder/derivatives/structural-provenance
recon-all -version > /your_folder/derivatives/structural-provenance/freesurfer_version.txt
conda env export --no-builds > /your_folder/derivatives/structural-provenance/structural_stats_environment.yml
cp /your_folder/BIDS/participants.tsv /your_folder/derivatives/structural-provenance/participants_used.tsv
cp /your_folder/code/structural/*.sh /your_folder/derivatives/structural-provenance/
cp /your_folder/code/structural/*.py /your_folder/derivatives/structural-provenance/
```

- Record the exact participant inclusion list and QC status.
- Preserve the original FreeSurfer logs.
- Keep scripts under version control.
- Do not manually edit exported tables without a scripted record of the change.
- Store statistical model formulas and contrasts as plain-text files.

20. Troubleshooting

Problem	Likely cause and action
recon-all: command not found	Source SetUpFreeSurfer.sh and verify FREESURFER_HOME.
License error	Set FS_LICENSE to a readable FreeSurfer license.txt.
Subject already exists	Do not reuse -i. Resume with recon-all -s subject -all or choose a new subject directory.
Job stopped after SSH disconnect	Run inside tmux; inspect the internal recon-all.log and resume.
Very poor surfaces	Check T1w motion, contrast, orientation, skull stripping, and scanner artifacts; follow a documented edit protocol.
asegstats2table skips a participant	Confirm the subject folder name and the existence of stats/aseg.stats.
FSGD subject not found	Input names must exactly match SUBJECTS_DIR folder names.
Freeview will not open remotely	Use X forwarding, a local workstation, or export QC snapshots on a graphical node.
Mixed versions detected	Stop and rebuild a consistent dataset; do not combine outputs silently.

21. Final checklist

- The T1w filename and participant label match.
- The exact FreeSurfer version and license path are recorded.
- recon-all.done exists for every included participant.
- White, pial, skull-strip, and aseg QC are complete and documented.
- Excluded subjects are removed by a scripted inclusion rule.
- ROI tables have correct units and one row per participant.
- Covariates and contrasts were prespecified.
- Multiple-comparison correction matches the inferential family.
- Scripts, environment files, logs, and participant lists are archived.

Appendix A. Included scripts

File	Purpose
01_check_freesurfer.sh	Verify installation, environment, license, and recon-all version.
02_run_recon_all.sh	Run one participant using BIDS defaults and preserve a log.
03_batch_recon_all.sh	Read participants.tsv and process participants sequentially or with controlled parallelism.
04_check_recon_status.sh	Interpret done, error, and incomplete states.
05_extract_roi_tables.sh	Export aseg and aparc volume, area, and thickness tables.
06_prepare_surface_glm.sh	Resample thickness to fsaverage and smooth it.
07_run_surface_glm.sh	Fit a vertex-wise FreeSurfer GLM.

File	Purpose
analyze_roi_stats.py	Fit ROI-wise OLS models and apply FDR.
environment.yml	Create the structural_stats Python environment.

```
cp -r /path/to/downloaded_bundle/scripts/structural /your_folder/code/
cd /your_folder/code/structural
chmod +x *.sh *.py
```

References

1. FreeSurfer software suite. <https://surfer.nmr.mgh.harvard.edu/>
2. FreeSurfer recon-all documentation. <https://surfer.nmr.mgh.harvard.edu/fswiki/recon-all>
3. FreeSurfer 7 release downloads and version information. <https://surfer.nmr.mgh.harvard.edu/fswiki/rel7downloads>
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9. Dale AM, Fischl B, Sereno MI. Cortical surface-based analysis I. NeuroImage. 1999;9(2):179-194.